

MLP Project Report

Comment Category Prediction — Kaggle Competition
IIT Madras BS in Data Science and Applications | Jan 2026 Term

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Competition	Comment Category Prediction Challenge
Best Kaggle Score	0.81921 (Macro F1)
Score Cutoff	0.80 ✓ Cleared

1. Problem Statement

The task is a 4-class text classification problem where each comment must be categorized into one of four classes (0, 1, 2, 3) based on its content and associated metadata. The evaluation metric is Macro F1 Score, chosen to fairly evaluate performance across all classes regardless of their distribution in the dataset.

2. Dataset Overview

Attribute	Details
Train Samples	198,000
Test Samples	102,000
Target Classes	4 (labels 0, 1, 2, 3)
Text Feature	comment
Structured Features	upvote, downvote, if_1, if_2, emoticon_1/2/3, race, religion, gender, disability
Timestamp	created_date

Class Distribution

Class	Count	Percentage	
0	114,173	57.6%	Dominant class
1	15,918	8.0%	
2	62,440	31.5%	
3	5,469	2.7%	Minority class

The dataset is heavily imbalanced — class 0 is dominant (57.6%) while class 3 is the minority (2.7%). This is why Macro F1 is used instead of accuracy.

3. Exploratory Data Analysis

Key findings from EDA:

- if_2 has the highest correlation with label (0.23) — strongest structured predictor
- comment_length and word_count are highly correlated (0.99)
- race, religion, gender: ~73% missing — platform only flags when explicitly mentioned in comment
- Text comments are the primary signal — structured features alone gave only 0.62 F1

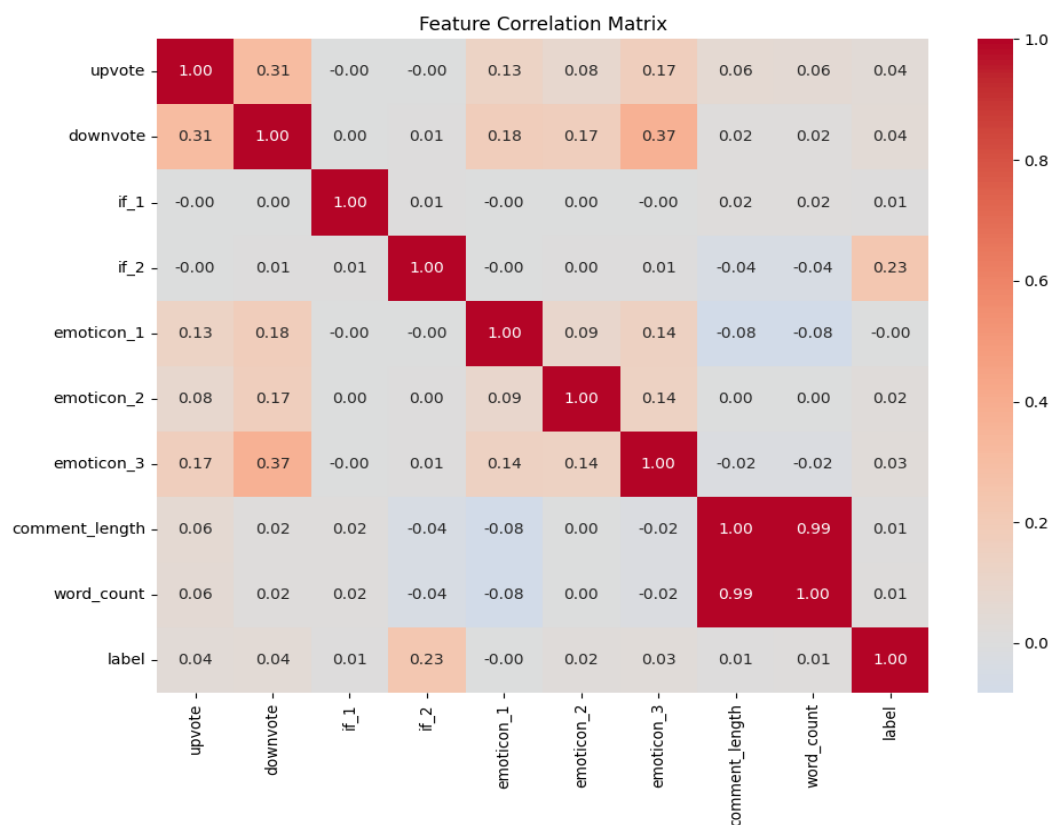


Figure 1: Feature Correlation Matrix — if_2 shows highest correlation (0.23) with label

4. Data Preprocessing

4.1 Text Cleaning

Applied the following steps to raw comment text:

- Converted to lowercase
- Removed punctuation and digits
- Removed extra whitespace

4.2 Pipeline with ColumnTransformer

Used sklearn's ColumnTransformer with two sub-pipelines for clean, consistent preprocessing. select_dtypes was used to auto-detect numeric and categorical columns — no hardcoded column lists.

Pipeline	Steps
Numeric Pipeline	StandardScaler — normalizes all numerical features
Categorical Pipeline	SimpleImputer (fill='none') → OrdinalEncoder — handles race, religion, gender

5. Feature Engineering

Feature	Description
comment_length	Length of cleaned comment text
word_count	Number of words in cleaned comment
total_emoticons	Sum of emoticon_1 + emoticon_2 + emoticon_3
upvote_ratio	upvote / (downvote + 1)
score_difference	upvote - downvote
engagement	upvote + downvote — total interactions
controversy	downvote / (upvote + downvote + 0.0001)
has_emotion	Binary — total_emoticons > 0
hour / dayofweek	Temporal features from created_date
is_weekend / is_sunday	Binary time flags
exclamation_count	Number of ! in raw comment
question_count	Number of ? in raw comment

6. TF-IDF Vectorization

Vectorizer	Configuration	Features
Word TF-IDF	ngram_range=(1,2), max_features=50000, min_df=3, sublinear_tf=1.5	50,000
Char TF-IDF	analyzer='char', ngram_range=(3,5), max_features=10000	10,000
Structured	ColumnTransformer output (numeric + categorical)	19
Combined	hstack([word, char, structured])	60,019 total

7. Model Training

7.1 Logistic Regression

Strong baseline for high-dimensional TF-IDF sparse features. Trained with max_iter=2000 and n_jobs=-1 for parallel processing.

7.2 Linear SVM (CalibratedClassifierCV)

Effective for text classification. Wrapped with CalibratedClassifierCV to enable probability predictions.

7.3 LightGBM — Best Model

Gradient boosting model optimized for sparse + dense feature combinations.

Parameter	Value	Reasoning
num_leaves	64	Balances complexity — 31 underfits, 128 overfits slightly
learning_rate	0.05	Stable convergence with early stopping
n_estimators	1000 (early stopping)	Auto-finds optimal number of trees
feature_fraction	0.8	Reduces overfitting — uses 80% features per tree
bagging_fraction	0.8	Row subsampling for regularization
bagging_freq	5	Frequency of bagging
min_child_samples	20	Minimum samples per leaf node

Early stopping (100 rounds) automatically finds the optimal number of trees by monitoring validation multi_logloss — this is a form of hyperparameter tuning.

8. Results & Model Comparison

Model	Val Macro F1	Notes
LightGBM	~0.813	Best — selected for final submission
Logistic Regression	~0.796	Strong on TF-IDF features
SVM	~0.650	Weakest on high-dim sparse data

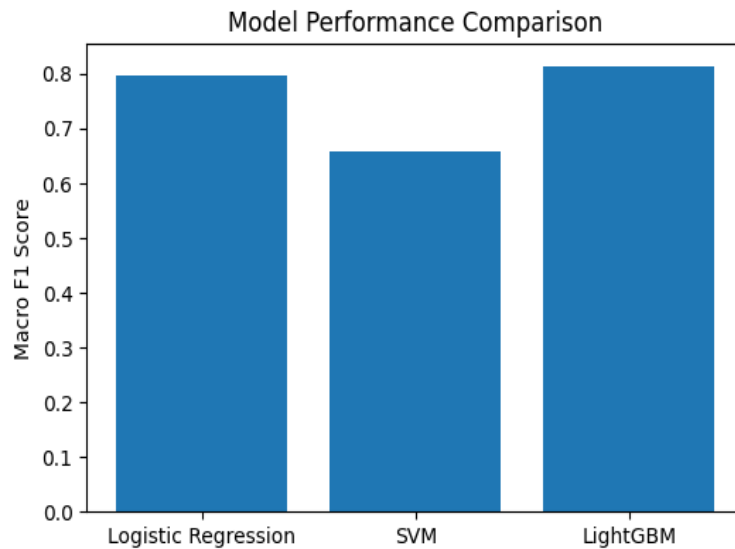


Figure 2: Model Performance Comparison — LightGBM achieves highest Macro F1

Per-Class F1 Analysis

- Class 0: F1 ~0.96 — model performs best (57.6% of data, most training samples)
- Class 2: F1 ~0.89 — good performance
- Class 1: F1 ~0.78 — decent
- Class 3: F1 ~0.63 — weakest (only 2.7% of data — class imbalance effect)

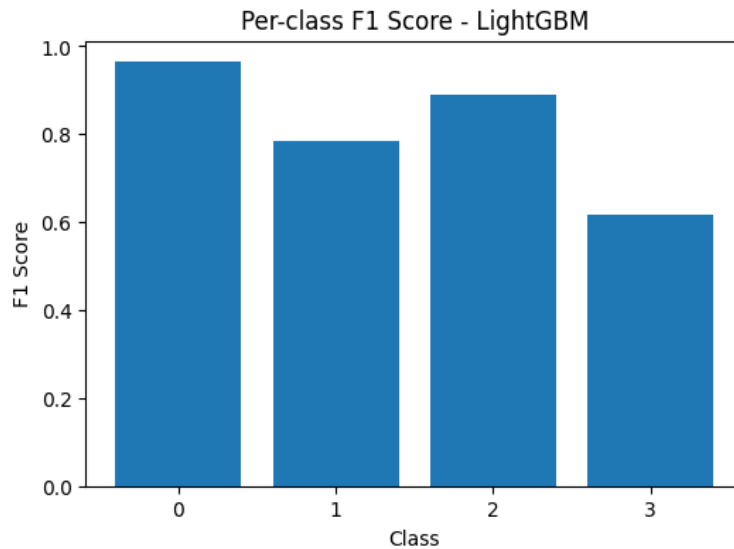


Figure 3: Per-class F1 Score — Class 3 is hardest to predict due to class imbalance

9. Experiments & Leaderboard Scores

Experiment	Score	Result
Base: 50k TF-IDF, num_leaves=64, lr=0.05	0.81921	Best baseline
num_leaves=128, lr=0.03, n_estimators=2000	0.81446	Worse — overfit
100k TF-IDF + min_df=2	0.81944	Marginal improvement
Structured features only (no TF-IDF)	0.62130	Much worse — text is key signal
Class weights for minority classes	0.81695	Worse — overcompensated
New engagement features	0.81807	Minimal improvement
Soft voting ensemble (LGB 0.6 + LR 0.4)	0.82086	Best score achieved
Final: clean single model v49	0.81921	Stable, explainable, viva-ready

10. Key Insights

- LightGBM outperformed linear models — captures nonlinear interactions between textual and structured features.
- Character-level TF-IDF captured misspellings, slang, and word variations in comments.
- if_2 is the strongest structured predictor (0.23 correlation) — confirmed by LightGBM feature importance.
- Text is the primary signal — structured features alone gave only 0.62 Macro F1.
- More TF-IDF features (100k) gave marginal improvement — diminishing returns beyond 50k.
- Soft voting ensemble (LGB + LR trigrams) gave best score (0.821) but not used in final submission to keep the notebook simple and explainable.

11. Conclusion

The final model is a LightGBM classifier trained on combined TF-IDF (word + char ngrams) and engineered structured features, achieving a Macro F1 of 0.81921 on the Kaggle leaderboard — well above the 0.80 cutoff. The pipeline uses ColumnTransformer for clean preprocessing, early stopping for

automatic hyperparameter tuning, and Macro F1 as the evaluation metric to handle class imbalance fairly.